

# Manal Abdulhakeem Bafaraj

Software Engineer (Full-Stack)

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## SUMMARY

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Full-Stack Developer with a strong foundation in front-end and backend development, experienced in building scalable web applications and managing databases. Skilled in technologies such as JavaScript, PostgreSQL, and NestJS, with hands-on experience in real-world projects and cloud environments. Passionate about creating efficient, user-focused solutions and continuously improving technical skills.

## EXPERIENCE

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10/2025 - 4/2026

### Software Engineer Trainee (Tamheer Program)

Institute of Hajj and Umrah Research – Umm Al-Qura University

- Participating in a 6-month Tamheer training program in software development.
- Developed and maintained web-based systems and internal tools, enhancing system performance and user experience.
- Worked on both front-end and back-end tasks, improving system performance and usability.
- Collaborated with teams using Agile methodologies and version control systems (Git).
- Gained hands-on experience in real-world development environments and problem-solving.

06/2025 - 9/2025

### Full Stack Web Development Bootcamp

Al-Khobar Experts Training Institute

- Completed intensive training in full-stack web development.
- Built responsive and accessible web interfaces using HTML, CSS, JavaScript, and Bootstrap.
- Developed practical projects focusing on modern UI/UX principles.
- Currently expanding knowledge in backend development and full-stack integration.

07/2025 - 10/2025

### Advanced UI/UX Design Bootcamp

Newtech Technical challenge

- Successfully completed a 3-month intensive training program in advanced UI/UX design.
- Gained hands-on experience in user interface design, user experience research, and prototyping.

07/2024 - 08/2024

### Software Engineer Intern (Full-Stack)

Digital Upgrade ME

- Developed and maintained dynamic web applications using HTML, CSS, JavaScript, Python, and PHP.
- Collaborated using Git and GitHub for version control and team-based development.
- Assisted in building and testing user interfaces and backend functionalities.
- Participated in debugging, code reviews, and improving system performance.
- Worked in an Agile environment using tools like Slack for task management and team communication.

## EDUCATION

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02/2020 - 03/2025

### Computer Science

Umm Al-Qura University

- Bachelor's Degree
- GPA: 3.66 / 4.00—Graduated with Honors
- Specialization: Machine Learning, Artificial Intelligence, and Data Science.
- Graduation Project: Speech Emotion Classification using Machine Learning and Deep Learning techniques—achieved 96.9% accuracy through MFCC and spectrogram-based feature extraction.

CERTIFICATES

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|---------|-----------------------------------------------------------|
| 04/2026 | <u>Supercharge Your Data Analytics with Generative AI</u> |
| 05/2025 | <u>Introduction to Cybersecurity</u>                      |
| 03/2025 | <u>Code Generation and Optimization Using IBM Granite</u> |
| 11/2024 | <u>Python Basics</u>                                      |
| 06/2024 | <u>Data Fundamentals</u>                                  |

SKILLS

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Technical Skills:

- Languages: JavaScript, Python, Java, C++
- Frontend: HTML, CSS, Bootstrap
- Backend: Node.js, NestJS
- Database: PostgreSQL, MySQL
- Tools: Git, GitHub, pgAdmin, VS Code
- Concepts: REST APIs, OOP, Database Design

Soft Skills:

- Teamwork and Collaboration
- Time Management
- Communication Skills
- Attention to Detail
- Self-learning, and Adaptability

LANGUAGES

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|--------|--------|---------|--------|
| Arabic | Native | English | Fluent |
|--------|--------|---------|--------|

PROJECTS

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| 11/2025 -03/2026 | <div>Full-Stack Developer</div> <div>Saden Project</div> <ul style="list-style-type: none"><li>• Developed a full-stack survey and field data collection system, including a mobile application for field data entry and a web-based admin dashboard for system management.</li><li>• Designed the mobile application for field workers (Observers) with offline support, enabling survey completion without internet connectivity and automatic data synchronization when connection is restored.</li><li>• Implemented geolocation tracking to capture the observer’s location during survey submission, improving data accuracy and reliability.</li><li>• Developed a web-based dashboard for Admin and Super Admin roles to manage users, surveys, and system permissions.</li><li>• Designed and implemented PostgreSQL database schemas with structured relationships and role-based access control (RBAC).</li><li>• Developed backend services and RESTful APIs using NestJS.</li><li>• Built a multi-role system including System Admin, Super Admin, Leader, and Worker with a strict and well-defined permission hierarchy.</li><li>• Enforced that only the System Admin can create Super Admin accounts, ensuring centralized control over the system.</li><li>• Implemented core features such as survey creation, assignment, submission, and tracking workflows.</li><li>• Integrated real-time functionalities including live location tracking for field teams.</li><li>• Participated in deployment, database migration, and system performance optimization.</li></ul> |
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05/2025 -06/2025

### Full-Stack Developer

Project: Online Exam Management System

[GitHub Repo](#)

- Developed a full-featured online exam platform for students and supervisors.
- Executed secure login, student profiles, and a real-time exam countdown timer.
- Enabled dynamic fetching of questions from the database and tracking of exam submissions.
- Designed a supervisor dashboard to efficiently manage exams, questions, and settings.
- Focused on usability, security, and a seamless user experience for both students and supervisors.

04/2025 -05/2025

### Front-End & Data Visualization Developer

Post-Graduation Project: Brute Force Attack Dashboard

[GitHub Repo](#) | [Live Demo](#)

- Built an interactive dashboard to analyze and visualize brute-force login attempts.
- Applied detection methods for suspicious activity, weak passwords, and time-based attack patterns.
- Integrated smart filters and dynamic metrics to explore uploaded Excel logs.
- Designed the dashboard following data visualization best practices to ensure clarity, usability, and actionable insights.

09/2024 -03/2025

### Machine Learning Developer

Graduation Project: Speech Emotion Classification Using Machine Learning

- Developed and trained two advanced machine learning models to classify six human emotions—happiness, sadness, anger, fear, disgust, and neutral—from speech signals.
- Utilized the CREMA-D dataset with over 7,400 diverse audio samples to ensure robust model accuracy and generalization.
- Applied key audio feature extraction techniques, including MFCCs, RMS, and Zero-Crossing Rate (ZCR).
- Engineered two Convolutional Neural Network (CNN) architectures—one using extracted features and another using spectrograms—achieving up to 96.9% accuracy.
- Enhanced model performance through data augmentation techniques such as noise addition, pitch shifting, and time stretching.
- Conducted a comparative analysis to demonstrate the impact of spectrograms on classification precision and overall model reliability.
- Documented results and insights in a formal research report, contributing to ongoing studies in Speech Emotion Recognition and Human-Computer Interaction.